OpenStack releases with the components included

OpenStack Austin - Nova, Swift
OpenStack Bexar - Nova, Glance, Swift
OpenStack Cactus - Nova, Glance, Swift
OpenStack Diablo - Nova, Glance, Swift
OpenStack Essex - Nova, Glance, Swift, Horizon, Keystone
OpenStack Folsom - Nova, Glance, Swift, Horizon, Keystone, Quantum, Cinder
OpenStack Grizzly - Nova, Glance, Swift, Horizon, Keystone, Quantum, Cinder
OpenStack Havana - Nova, Glance, Swift, Horizon, Keystone, Neutron, Cinder, Heat, Ceilometer
OpenStack Icehouse - Nova, Glance, Swift, Horizon, Keystone, Neutron, Cinder, Heat, Ceilometer, Trove

OpenStack computing components

OpenStack has a modular architecture that controls large pools of compute, storage and networking resources.

Compute (Nova): OpenStack Compute (Nova) is the fabric controller, a major component of Infrastructure as a Service (IaaS), and has been developed to manage and automate pools of computer resources. It works in association with a range of virtualisation technologies. It is written in Python and uses many external libraries such as Eventlet, Kombu and SQLAlchemy.

Object storage (Swift): It is a scalable redundant storage system, using which objects and files are placed on multiple disks throughout servers in the data centre, with the OpenStack software responsible for ensuring data replication and integrity across the cluster. OpenStack Swift replicates the content from other active nodes to new locations in the cluster in case of server or disk failure.

Block storage (Cinder): OpenStack block storage (Cinder) is used to incorporate continual block-level storage devices for usage with OpenStack compute instances. The block storage system of OpenStack is used to manage the creation, mounting and unmounting of the block devices to servers. Block storage is integrated for performance-aware scenarios including database storage, expandable file systems or providing a server with access to raw block level storage. Snapshot management in OpenStack provides the authoritative functions and modules for the back-up of data on block storage volumes. The snapshots can be restored and used again to create a new block storage volume.

Networking (Neutron): Formerly known as Quantum, Neutron is a specialised component of OpenStack for managing networks as well as network IP addresses. OpenStack networking makes sure that the network does not face bottlenecks or any complexity issues in cloud deployment. It provides the users continuous self-service capabilities in the network's infrastructure. The floating IP addresses allow traffic to be dynamically routed again to any resources in the IT infrastructure, and therefore the users can redirect traffic during maintenance or in case of any failure. Cloud users can create their own networks and control traffic along with the connection of servers and devices to one or more networks. With this component, OpenStack delivers the extension framework that can be implemented for managing additional network services including intrusion detection systems (IDS), load balancing, firewalls, virtual private networks (VPN) and many others.

Figure 1: OpenStack

Dashboard (Horizon): The OpenStack dashboard (Horizon) provides the GUI (Graphical User Interface) for the access, provision and automation of cloud-based resources. It embeds various third party products and services including advance monitoring, billing and various management tools.

Identity services (Keystone): Keystone provides a central directory of the users, which is mapped to the OpenStack services they are allowed to access. It refers and acts as the centralised authentication system across the cloud operating system and can be integrated with directory services like LDAP. Keystone supports various authentication types including classical username and password credentials, token-based systems and other log-in management systems.

Image services (Glance): OpenStack Image Service (Glance) integrates the registration, discovery and delivery services for disk and server images. These stored images can be used as templates. It can also be used to store and catalogue an unlimited number of backups. Glance can store disk and server images in different types and varieties of back-ends, including Object Storage.

Telemetry (Ceilometer): OpenStack telemetry services (Ceilometer) include a single point of contact for the billing systems. These provide all the counters needed to integrate customer billing across all current and future OpenStack components.

Orchestration (Heat): Heat organises a number of cloud applications using templates with the help of the OpenStack-native REST API and a CloudFormation-compatible Query API.

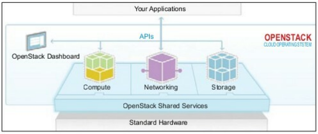

Figure 2: OpenStack: an open source cloud operating system
[Source: openstack.org]